

# Dr. Georgii Oblapenko

Born on 10 February 1992

Chair of Applied & Computational Mathematics (ACoM), RWTH Aachen, Schinkelstraße 2, 52062 Aachen, Germany



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Languages: English (fluent), German (fluent, including technical German), Russian (native speaker)

Researcher with 10+ years of experience in the field of numerical model development for simulation of non-equilibrium flows.

## Employment history

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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nov. 2023 – Present    |  <b>Member of Scientific Staff</b> • RWTH Aachen, Chair of Applied and Computational Mathematics.<br>Development of kinetic methods for modelling of non-equilibrium flows in continuum and rarefied regimes. Development of entropy-stable DG schemes for multi-component real gas flows with thermo-chemical non-equilibrium. Teaching lectures and global exercises for Bachelor and Master students.                                                                                                                         |
| Jan. 2023 – Sep. 2023  |  <b>Scientific Consultant</b> • UT Austin, Oden Institute for Computational Engineering and Sciences / Department of Department of Aerospace Engineering and Engineering Mechanics.<br>Development of new PIC-DSMC methods. Co-supervision of Master student developing PIC-DSMC methods for high-voltage gap closure modeling.                                                                                                                                                                                                |
| Nov. 2021 – Oct. 2023  |  <b>Postdoctoral Researcher</b> • German Aerospace Center (DLR), Institute of Aerodynamics and Flow Technology, Spacecraft Department.<br>Uncertainty quantification of hypersonic expanding continuum flows. Development of an efficient photon Monte Carlo radiation transport code. Development of novel kinetic methods for simulations of plasma and gases with internal degrees of freedom. Work on low-rank methods for the Boltzmann equation. Supported by a personal grant of the Alexander von Humboldt Foundation. |
| July 2018 – June 2021  |  <b>Postdoctoral Research Fellow</b> • UT Austin, Oden Institute for Computational Engineering and Sciences / Department of Department of Aerospace Engineering and Engineering Mechanics.<br>Development of hybrid Discrete Velocity Method/Direct Simulation Monte Carlo methods. Development of new Direct Simulation Monte Carlo algorithms for plasma simulations. Development of new kinetic models for gases with internal degrees of freedom and chemically reacting gas flows.                                        |
| Sept. 2013 – Dec. 2017 |  <b>Research engineer</b> • Saint-Petersburg State University, Department of Hydroaeromechanics.<br>Lead developer of a C++ library for kinetic theory computations and coupling with CFD codes. Development of theoretical models of reaction rates in viscous gas flows and their numerical modeling. Development of simplified models for vibrational relaxation rates.                                                                                                                                                     |

## Education

- 2015 – 2017     **Ph.D., Saint-Petersburg State University** in Physical and Mathematical Sciences.  
Area of specialization: Mechanics of Fluids, Gases and Plasma.  
Thesis title: *Physico-chemical relaxation rates in viscous non-equilibrium gas flows*.  
Supervisor: Prof. Kustova E.V.
- 2013 – 2015     **M.Sc. (summa cum laude), Saint-Petersburg State University** in Mechanics and Mathematical Modeling.  
Area of specialization: Molecular-kinetic theory of fluids and gases.  
Thesis title: *Vibrational relaxation rate in viscous rarefied gas flows*.  
Supervisor: Prof. Kustova E.V.
- 2009 – 2013     **B.Sc., Saint-Petersburg State University** in Mathematics and Mechanics.  
Supervisor: Prof. Kustova E.V.

## Awards and personal funding

### Personal funding and travel grants

- 2025     **DFG individual grant**, under review (re-submission). Application for funding of a doctoral position for 3 years. Initial review: research idea rated as very good, revision based on reviewer comments on the literature review. (Approx. 270000 EUR)
- 2021 – 2023     **Alexander von Humboldt Foundation** Research Stipend for Postdoctoral Researchers (60000 EUR).
- 2017 – 2018     **German Academic Exchange Service (DAAD)** Research Stipend (3300 EUR).
- 2016     **Santander Bank** Research Stipend (1400 EUR).
- 2015 – 2016     **Russian Foundation for Basic Research** Research Stipend (4500 EUR).
- 2013, 2014, 2017     **Saint-Petersburg State University** Travel funding awards.

### Awards

- 2023     **Marie Skłodowska-Curie Postdoctoral Action Seal of Excellence**
- 2017     **Stipend** of the Russian President for students and PhD students in prioritized study programmes (3800 EUR).
- 2016     **Stipend** of the Russian Government for students and PhD students in prioritized study programmes (1900 EUR).
-  **Stipend** of the Russian Government for students and PhD students (500 EUR).
- 2013     **Winner** of the Saint-Petersburg Government grant competition for students and graduate students (500 EUR).

## Publications

The full list of publications consists of **31** (including accepted papers) peer-reviewed publications in SCOPUS/Web of Science-indexed journals and conference proceedings, as well as **19** publications without peer review (conference proceedings, preprints).

Publication statistics (as of December 2025):

|                        | SCOPUS | Google Scholar |
|------------------------|--------|----------------|
| Number of publications | 29     | 58             |
| h-index                | 11     | 13             |
| Citations              | 359    | 587            |

## Conference and seminar presentations

|                       |   |                                                                                                                   |
|-----------------------|---|-------------------------------------------------------------------------------------------------------------------|
| Conferences           | ■ | Presented at <b>25</b> international and <b>11</b> national (German, Russian) conferences.                        |
| Plenary talks         | ■ | <b>2</b> invited plenary lectures at international conferences and workshops                                      |
| Invited seminar talks | ■ | <b>25</b> invited talks at universities and research institutes in the USA, Germany, France, and the Netherlands. |

## Teaching and student supervision

### Teaching

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|---------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summer semester 2026      | ■ | (Upcoming) Development and teaching of new Master course on “Modern Numerical Software Development”, RWTH Aachen (together with Dr. Lambert Theisen). Course offered to Master and Bachelor students of the Computational Engineering Science and Simulation Sciences programs. |
| Winter semester 2025/2026 | ■ | Global Exercise for “Foundations of Mathematics I”, Bachelor Study Program “Computational Engineering Science”, RWTH Aachen.<br>Invited lecture on kinetic theory in the framework of the Master course on “Aerospace Fundamentals”, Justus Liebig University Gießen.           |
| Summer semester 2025      | ■ | Development and teaching of new Master course on “Particle-based Simulation Methods”, RWTH Aachen. Course offered to Master and Bachelor students of the Mathematics, Computational Engineering Science, and Simulation Sciences programs.                                      |
| Winter semester 2024/2025 | ■ | Invited lecture on kinetic theory in the framework of the Master course on “Aerospace Fundamentals”, Justus Liebig University Gießen.                                                                                                                                           |
| Summer semester 2024      | ■ | Global Exercise for “Foundations of Mathematics II”, Bachelor Study Program “Computational Engineering Science”, RWTH Aachen.                                                                                                                                                   |
| Winter semester 2023/2024 | ■ | Global Exercise for “Foundations of Mathematics I”, Bachelor Study Program “Computational Engineering Science”, RWTH Aachen.<br>Invited lecture on kinetic theory in the framework of the Master course on “Aerospace Fundamentals”, Justus Liebig University Gießen.           |

### Additional teaching qualifications

|           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025      | ■ | “Basics Module” certificate of the program “Professionelle Lehrkompetenz für die Hochschule”, Hochschuldidaktik NRW                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2024–2025 | ■ | Completed courses at the Center for Teaching and Learning Services, RWTH Aachen:<br>Academic Teaching ◦ Lecture Planning ◦ Fundamentals of Examination<br>◦ Mündliche Prüfung - mehr als ein Gespräch?! ◦ Media Competency in Teaching<br>◦ Evaluation und Selbstreflexion in meiner Lehre ◦ Lernförderliche Beratung von Studierenden für Prüfung und Klausur<br>◦ Studien- und Abschlussarbeiten betreuen ◦ Unconscious Bias and Intercultural Training<br>◦ Dynexite: e-Prüfung planen und durchführen ◦ Basics of RWTH Moodle. |
| 2020      | ■ | Completed Title IX Training on prevention and reporting of sexual discrimination and harassment, UT Austin.                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Supervision, thesis reviews

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|----------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 – present | ■ | Regular supervision of Bachelor theses at RWTH Aachen.<br>Supervision of seminar and group research projects at RWTH Aachen.<br>Referee of Bachelor theses. |
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## Teaching and student supervision (continued)

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|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 – present | Regular supervision of student research assistants at RWTH Aachen.<br>Co-supervision of PhD Student Eda Yilmaz.                                  |
| 2022 – 2023    | Co-supervision of a Master thesis at UT Austin.                                                                                                  |
| 2020 – 2021    | Referee of Bachelor-, Diploma- and Master theses at Saint-Petersburg State University and Peter the Great St. Petersburg Polytechnic University. |

## Professional membership and other activities

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|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 – 2026             | <b>RWTH hiring committee</b> • Deputy member of the hiring committee for the W2 professorship “Industrial mathematics”, RWTH Aachen.                                                                           |
| Mar. 2022 – Mar. 2024   | <b>COST Network</b> • Member of the “Mathematical models for interacting dynamics on networks” (MAT-DYN-NET) networking group, supported by the COST (European Cooperation in Science and Technology) program. |
| Jul. 2021 – Present     | <b>RGD NextGen Group</b> • Member of the RGD NextGen Group, aimed at increasing outreach of the Rarefied Gas Dynamics community.                                                                               |
| Spring 2021 – Jan. 2024 | <b>American Institute of Aeronautics and Astronautics (AIAA) Thermophysics Technical Committee</b> • Member of Education, Emerging Technologies, Conference subcommittees.                                     |
| Dec. 2019 – Jan. 2024   | <b>American Institute of Aeronautics and Astronautics (AIAA)</b> • Senior AIAA Member.                                                                                                                         |

## Editing and review activities

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|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Editor                              | Proceedings of the 33rd International Symposium on Rarefied Gas Dynamics, Springer, 2026 (co-editor)                                                                                                                                                                                                                                                                           |
| Journal reviewer                    | Computers and Fluids (Elsevier) · Acta Astronautica (Elsevier) · Journal of Computational Physics (Elsevier) · Computer Methods in Applied Mechanics and Engineering (Elsevier) · Applied Mathematical Modelling (Elsevier) · Physics of Fluids (AIP) · Physics of Plasmas (AIP) · Journal of Thermophysics and Heat Transfer (AIAA) · Journal on Scientific Computing (SIAM). |
| Proceedings and conference reviewer | International Symposium on Rarefied Gas Dynamics · STAB/DGLR Symposium · AIAA Scitech · AIAA Aviation                                                                                                                                                                                                                                                                          |

## Event organisation

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|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RGD33 Symposium         | Member of the Local Organizing Committee of the “33rd International Symposium on Rarefied Gas Dynamics”, 15-19 July 2024, Göttingen, Germany.                          |
| VIII Polyakhov Readings | Member of the Local Organizing Committee of the international conference on mechanics “VIII Polyakhov Readings”, 30 January-2 February 2018, Saint-Petersburg, Russia. |

## Other activities

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- 2025      Online Pop-Science Talk “Die Offenbarung des Teilchens / The revelation of the particle” as part of “Falling Rhythm” exhibition of the “o43” artist collective, Barcelona, Spain.
- 2016 – **present**      Co-founder of the new media collective “GODA”. Complete development (concept, graphic design, coding, music composition) of interactive art objects and art installations.
- 2015-2016      Music studies • Sound Museum, Sain-Petersburg, Russian Federation.  
Improvised new music, graphical notation.